Monocular Depth Estimation as a Domain-Invariant Representation for Vision-Based Quadruped Parkour

Abstract

Parkour poses a significant challenge for legged robots, requiring navigation through complex environments with agility and precision based on limited sensory inputs. Vision-based quadruped parkour typically relies on onboard depth cameras, limiting deployment to robots with specialized sensors. In this work, we investigate whether a pre-trained monocular depth estimator can serve as a domain-invariant bridge, enabling sim-to-real transfer from only an RGB camera. We present a two-phase pipeline: a privileged teacher trained via PPO with ground-truth terrain scans, distilled into a visual student via DAgger operating on depth images from Depth Anything V2 (DA2). During simulated evaluations, we demonstrate that this approach results in limited emergent parkour capabilities, under specific conditions defined by a *Depth Signal Quality* metric.

1 Introduction

In recent years, training locomotion policies in simulation using reinforcement learning (RL) then transferring them to real robots has proven highly effective in mastering agile skills [15, 2, 8, 18, 22, 3, 16, 10, 12, 17]. Notably, [4, 28, 6, 11] have successfully demonstrated a range of dynamic skills akin to the athletic maneuvers seen in parkour, including walking, climbing, jumping, and crawling, which necessitates tight coordination between vision and control.

Agile locomotion over challenging terrain is a central goal of legged robotics. Recent work has shown that RL combined with teacher–student distillation can produce quadruped controllers capable of remarkable parkour [6, 28, 5]. These systems follow a two-phase paradigm: a *teacher* trains with privileged terrain information, then a *student* reproduces the teacher's behavior from realistic sensors.

A critical assumption is the availability of an onboard depth sensor. Intel RealSense or structured-light cameras provide dense depth that transfers well from simulation. However, not all platforms include depth sensors—the Unitree Go2 ships with only a front-facing RGB camera, precluding direct application of depth-based parkour methods.

Yet, training visual locomotion policies presents significant challenges due to the limited field of view and dynamic, lag-prone visual perception during dynamic movements. Previous methods for end-to-end control from pixels [1, 28, 6] typically involve a two-stage process, where a policy is first trained with privileged information about the robot's surroundings, then the learned behaviors are distilled into a visual policy using imitation learning [21]. However, such an approach relies on the unrealistic assumption that privileged information can be fully reconstructed from a history of depth images. Indeed, such privileges may reveal information obscured behind obstacles, outside the field of view of the egocentric camera, or simply missing from the depth image stream due to lags. This gap between the privileged information and the actual visual inputs may result in behaviors that a vision-based policy cannot replicate accurately. In that case, an optimal vision policy would differ from the one using privileged information. It would, for example, try to gather more information before crossing an obstacle to handle the occlusion. Ideally, training RL directly from pixels would ensure the policy learns behaviors effectively based on its actual visual sensors, but this method is impractical due to the high computational costs associated with generating depth images in simulations [1].

Deploying vision-based policies trained in simulation requires bridging the *sim-to-real gap*. Three strategies have emerged: (1) **domain randomization** [25], which randomizes visual properties; (2) **generative augmentation** [27], which uses diffusion models for photorealistic images; and (3) **canonical representations**, which project both domains into a shared space via a foundation model.

We pursue strategy (3) using Depth Anything V2 (DA2) [26]. If DA2 produces structurally consistent depth from both simulated and real RGB, its output constitutes a domain-invariant canonical representation. A student trained on DA2 depth in simulation deploys on DA2 depth from real images without adaptation—the policy never sees raw RGB. This is simpler than generative approaches and more principled than domain randomization.

Our contributions:

- We demonstrate monocular depth as a sim-to-real bridge for quadruped parkour, achieving 94–100% on stairs, 96–99% on hurdles, and 80–96% on gaps (within 7 pp of GT depth).
- A complete pipeline integrating DA2 into teacher–student distillation with a recurrent depth encoder and windowed BPTT.
- Systematic ablation (30+ experiments) identifying depth model capacity and terrain texture as dominant factors.
- A Depth Signal Quality metric diagnosing the failure mode on textureless surfaces.

2 Related Work

RL for quadruped locomotion. PPO [24] is the standard for training locomotion in simulation [22]. The teacher–student paradigm was popularized by RMA [14].

Quadruped parkour. Cheng et al. [6] demonstrated extreme parkour on a Go1 using a scandot teacher distilled into a depth-camera student via DAgger—our architectural basis. Chane-Sane et al. [5] introduced SoloParkour with constrained MDPs (CaT) and off-policy DDPG distillation. Zhuang et al. [28] present iterative distillation with depth perception. All three require onboard depth sensors.

Sim-to-real for visual policies. Domain randomization [25] provides no coverage guarantees. LucidSim [27] generates photorealistic images via Stable Diffusion conditioned on simulation depth, but requires running a diffusion model during training. Our approach uses a single foundation model call to create a shared representation space.

Monocular depth estimation. From MiDaS [20] to DA2 [26], monocular depth models now achieve remarkable zero-shot generalization. DA2 uses a DINOv2 [19] encoder with metric variants at Small (25M), Base (100M), and Large (300M) scales. Its application to agile locomotion control is largely unexplored.

3 Method

3.1 Problem Formulation

We formulate quadruped parkour as an MDP $\mathscr{M} = \langle \mathscr{S}, \mathscr{A}, \mathscr{T}, R, \gamma \rangle$. The action space $\mathscr{A} = \mathbb{R}^{12}$ consists of joint position offsets, converted to targets via $q_{\text{target}} = q_0 + \alpha a_t$ ($\alpha = 0.25$). A PD controller computes torques:

$$\tau = K_p(q_{\text{target}} - q) - K_d \dot{q}, \tag{1}$$

with $K_p = 30$ Nm/rad and $K_d = 0.75$ Nms/rad (subject to randomization). Simulation runs at 200 Hz; the policy at 50 Hz (decimation 4), $\gamma = 0.998$.

Observations. All policies receive proprioception $o_t^{\text{prop}} \in \mathbb{R}^{53}$:

$$o_t^{\text{prop}} = [\underbrace{g_t}_{7}, \underbrace{R_t^\top \hat{g}}_{3}, \underbrace{\omega_t}_{3}, \underbrace{(q_t - q_0)s_q}_{12}, \underbrace{\dot{q}_t s_{\dot{q}}}_{12}, \underbrace{a_{t-1}}_{12}, \underbrace{c_t}_{4}], \tag{2}$$

where g_t contains goal displacements, $R_t^\top \hat{g}$ is projected gravity, ω_t is angular velocity (scaled 0.25), $q_t, \dot{q}_t$ are joint positions/velocities (scales s_q=1, $s_{\dot{q}}$=0.05), a_{t-1} is the previous action, and $c_t \in \{0,1\}^4$ are foot contacts.

The teacher additionally receives a scandot heightmap $o_t^{\text{scan}} \in \mathbb{R}^{132}$: heights at 132 points on a 12×11 grid spanning $[-0.3, 1.35]$ m forward and ± 0.75 m laterally. The student observes a depth image $d_t \in \mathbb{R}^{1 \times 58 \times 87}$ from an 87° HFOV camera.

Table 1: Obstacle parameters across difficulty rows.

Row	#Obs	Stair h	Hurdle h	Gap w
0	2	6 cm	10 cm	24 cm
1	3	8 cm	14 cm	32 cm
2	3	10 cm	20 cm	42 cm
3	3	12 cm	26 cm	50 cm

Rewards. The reward combines goal-tracking with regularization:

$$R(s_t, a_t) = \sum_k w_k \cdot r_k(s_t, a_t) \cdot \Delta t, \tag{3}$$

with $\Delta t = 0.02$ s. Primary terms include velocity tracking (w=3.0, exponential), heading (w=1.5), progress (w=2.0), waypoint bonuses (+20), and course completion (+40). Penalties cover collisions, torques, action rate, and orientation, with terrain-conditioned scaling ($\beta = 0.35$) relaxing penalties during jumps. The complete table is in Appendix A.

Terrain curriculum. A 4×12 grid: 4 difficulty rows $\times$ 12 columns (3 families—stairs, hurdles, gaps—$\times$ 4 variants). Table 1 gives obstacle parameters. Difficulty advances when progress > 0.85, regresses when < 0.40.

Domain randomization. Friction $\mu \sim \mathscr{U}[0.2, 1.7]$, added mass $\sim \mathscr{U}[-1,1]$ kg, CoM offset $\sim \mathscr{U}[-0.03, 0.03]^3$ m, PD gain scales $\sim \mathscr{U}[0.8, 1.2]$, and external pushes $\sim \mathscr{U}[-1,1]^2$ m/s every 15 s.

3.2 Privileged Teacher

The teacher π_ϕ uses a scandot encoder $f_\phi : \mathbb{R}^{132} \to \mathbb{R}^{32}$ (layers $128 \to 64 \to 32$, ELU) and an actor MLP:

$$\pi_\phi(o_t) = \text{Hardtanh}\big(\text{MLP}_{[256,256,128,12]}([o_t^{\text{prop}}; f_\phi(o_t^{\text{scan}})])\big). \tag{4}$$

The critic $V_\psi : \mathbb{R}^{232} \to \mathbb{R}$ receives an augmented observation with velocity, randomization parameters, and contacts.

Training uses PPO [24]:

$$\mathscr{L}_{\text{PPO}} = -\mathbb{E}_t\Big[\min\big(\rho_t \hat{A}_t, \text{clip}(\rho_t, 1 \pm \varepsilon)\hat{A}_t\big)\Big] + c_v \mathscr{L}_V - c_e H[\pi], \tag{5}$$

where $\varepsilon = 0.2$, $c_v = 1.0$, $c_e = 0.01$, and $\hat{A}_t$ is the GAE [23]:

$$\hat{A}_t = \textstyle\sum_{l=0}^{T-t} (\gamma\lambda)^l \delta_{t+l}, \quad \delta_t = r_t + \gamma V(s_{t+1}) - V(s_t), \tag{6}$$

with $\lambda = 0.95$. Training: 4096 environments, 48 steps/update, Adam [13] (lr=10^{-3}), 2500 iterations (~2 h on L40S).

3.3 Visual Student Distillation

The student replaces scandots with a depth image $d_t \in \mathbb{R}^{1 \times 58 \times 87}$. A recurrent encoder g_θ processes the depth stream.

CNN backbone.

$$d_t \xrightarrow{\text{Conv}_{5\times5}^{1\to32}} \xrightarrow{\text{Pool}_2} \xrightarrow{\text{Conv}_{3\times3}^{32\to64}} \xrightarrow{\text{FC}} z_{\text{cnn}} \in \mathbb{R}^{32}. \tag{7}$$

The FC layers map 62,400 → 128 → 32 (flattening the 64×25×39 feature map). All activations use ELU.

Temporal fusion. CNN features are concatenated with masked proprioception (heading zeroed) and processed through a GRU [7]:

$$\begin{aligned} z_{\text{combo}} &= \text{MLP}_{[128,32]}\left([z_{\text{cnn}}; \tilde{o}_t^{\text{prop}}]\right), \\ h_t &= \text{GRU}(z_{\text{combo}}, h_{t-1}) \in \mathbb{R}^{512}, \\ [z_t, \hat{\psi}_t] &= \tanh(W_{\text{out}} h_t + b_{\text{out}}) \in \mathbb{R}^{34}, \end{aligned} \tag{8}$$

yielding depth latent $z_t \in \mathbb{R}^{32}$ and predicted heading $\hat{\psi}_t \in \mathbb{R}^2$.

Actor. Same MLP as the teacher, *initialized with teacher weights*: $\pi_\theta = \text{Hardtanh}(\text{MLP}([o_t^{\text{prop}}; z_t]))$.

DAgger distillation. The student executes its own actions while the teacher provides labels [21]:

$$\mathcal{L} = \underbrace{\|a_t^{\text{teach}} - a_t^{\text{stud}}\|_2}_{\text{action}} + \underbrace{\|\psi_t^{\text{oracle}} - \hat{\psi}_t\|_2}_{\text{yaw}}. \tag{9}$$

Windowed BPTT. Full BPTT over 120 steps is infeasible (DA2 consumes ∼11 GB). We partition into K=5 windows of W=24 steps:

$$\nabla_\theta \mathcal{L} \approx \frac{1}{K} \sum_{k=1}^{K} \nabla_\theta \sum_{t=kW}^{(k+1)W-1} \mathcal{L}_t. \tag{10}$$

Hidden states are detached at window boundaries. Each window spans ∼5 depth frames (depth at ∼10 Hz). Gradient clipping at $\|\nabla_\theta\| \leq 1.0$.

MTS yaw gating. Mixture-of-Teacher-Student gating [6]:

$$o_t^{\text{heading}} = \begin{cases} 1.5 \cdot \hat{\psi}_t & \text{if } \|\hat{\psi}_t - \psi_t^{\text{oracle}}\| < 0.6, \\ \psi_t^{\text{oracle}} & \text{otherwise,} \end{cases} \tag{11}$$

injecting student heading only when close to the oracle.

Training recipe. Two critical choices: (1) differential LR—encoder uses $\eta = 2\times10^{-3}$, actor uses 0.1η; (2) cosine annealing:

$$\eta(t) = \eta_{\min} + \tfrac{1}{2}(\eta_0 - \eta_{\min})(1 + \cos(\pi t / T_{\max})), \tag{12}$$

with $\eta_{\min} = 10^{-4}$, $T_{\max} = 5000$. Training: 7000 iterations, 48 environments (∼30 h on L40S).

3.4 Canonical Depth via Monocular Estimation

At each depth update (∼10 Hz), rendered RGB is passed through DA2: $\hat{d}_t = \text{DA2}_{\text{metric-base}}(I_t)$ (∼100M params, FP16).

Table 2: Baseline gap success (%, 256 episodes/row).

Policy	Gap width 0.24 m	0.32 m	0.42 m	0.50 m
Teacher	98.5	100.0	99.0	98.5
GT-depth	94.5	94.5	89.1	87.5

EMA percentile normalization. DA2's raw output exhibits scale offsets on synthetic renders. We normalize via exponential moving average bounds:

$$\begin{aligned} \hat{q}_{\text{lo}}^{(t)} &= (1-\alpha)\hat{q}_{\text{lo}}^{(t-1)} + \alpha \cdot Q_{0.02}(\hat{d}_t), \\ d_t^{\text{norm}} &= \text{clip}\left(\frac{\hat{d}_t - \hat{q}_{\text{lo}}}{\hat{q}_{\text{hi}} - \hat{q}_{\text{lo}}}, 0, 1\right) - 0.5, \end{aligned} \tag{13}$$

with $\alpha = 0.02$, output in $[-0.5, 0.5]$.

Depth Signal Quality (DSQ). To diagnose depth information quality:

$$\text{DSQ} = \frac{|\bar{d}_{\text{obstacle}} - \bar{d}_{\text{ground}}|}{\sigma_d}. \tag{14}$$

On textureless terrain: Pearson $r = -0.10$—effectively zero. DA2 cannot infer depth from featureless surfaces. Adding checkerboard texture (UV scale 10.0) provides monocular cues that dramatically improve the signal.

4 Experiments

4.1 Setup

We use Genesis with GPU-accelerated simulation and batch CUDA rendering on NVIDIA L40S (45 GB). DA2 limits the student to 48 parallel environments. Evaluation uses *forced-row evaluation*: curriculum disabled, 256 episodes per condition. Primary metric: success rate (fraction completing all waypoints).

4.2 Baselines

Table 2 establishes bounds. The teacher achieves near-perfect success; the GT-depth student 87–95%, the architecture's upper bound.

4.3 Main Results

Our best configuration—DA2-base (100M) with checkerboard texture, differential LR, cosine annealing—achieves the results in Table 3 at checkpoint 7000. The student performs strongly across all families: 94–99% stairs, 96–99% hurdles, 80–96% gaps. Even on the hardest row (0.50 m gaps, ∼1.5× body length), success reaches 80–91%. Two gap evaluation runs illustrate terrain seed variance.

4.4 Ablation Study

We conducted 30+ experiments across four phases. Table 4 summarizes key findings on gap Row 3 (0.50 m, hardest).

Table 3: DA2 student success (%) across obstacle families (256 episodes/row). Gaps show two runs for variance.

	Difficulty row			
Family	R0	R1	R2	R3
Stairs (6–12 cm)	99.2	99.6	97.3	94.5
Hurdles (10–26 cm)	98.8	96.5	99.2	96.5
Gaps Run 1 (24–50 cm)	87.5	93.0	87.1	80.5
Gaps Run 2	94.1	96.1	94.9	91.0
GT-depth (gaps)	94.5	94.5	89.1	87.5

Table 4: Ablation on gap Row 3 (0.50 m).

Factor	**Config**	**%**
Model size	DA2-small (25M)	1–8
	DA2-base (100M)	**80–91**
Texture	None + DA2-base	1
	Checkerboard	**80–91**
Temporal	No smoothing	**80–91**
	EMA (α=0.5)	60–72
Normalization	EMA percentile	**80–91**
	Calibrated linear	$\sim$78
	Fixed range	$\sim$75
	Mean-std	$\sim$77
Training	Diff-LR + cosine	**80–91**
	Uniform LR	30–48
	Frozen actor	<20

Depth model capacity is the dominant factor: DA2-base improves Row 0 from 56% to 68% and Row 1 from 34% to 83% over DA2-small.

Terrain texture is the critical enabler: without it, DA2-base achieves $\sim$1% on Row 3; checkerboard raises this to 80–91%, consistent with DSQ analysis.

Temporal smoothing is counterproductive: EMA blurs the sharp depth discontinuities at obstacle edges.

Normalization is secondary: all four schemes produce comparable results once signal quality is adequate.

Training recipe is essential: differential LR prevents the warm-started actor from destabilizing before the encoder converges.

4.5 Zero-Depth Ablation

Replacing depth with zeros at inference: gap Row 3 drops from 80–91% to 0.0%, progress ratio from $\sim$0.95 to $\sim$0.35 (walks forward but cannot detect gaps).

4.6 Depth Signal Quality Analysis

Per-pixel correlation between DA2 and GT terrain over 2000 steps: on textureless terrain, $r = -0.10$; gap-ground depth difference at 4.5% of normalized range. EMA bounds stabilize at $q_{0.02} = 4.93$ m, $q_{0.98} = 11.49$ m (span 6.56 m), within which a 0.50 m gap is negligible. With checkerboard texture, terrain boundaries create edges that DA2 interprets as depth discontinuities, substantially increasing correlation.

4.7 Direct RGB Baseline

A frozen ResNet-18 [9] backbone (conv1–layer2, 683K params) on raw RGB achieved 98% overall and 69% on gaps, training 5× faster (3.2 s/iter, 2048 envs, 16 GB VRAM). However, ImageNet features lack the domain-invariance guarantee of canonical depth.

5 Discussion

Canonical depth occupies a middle ground: it requires only an RGB camera (like LucidSim [27]) but avoids generative pipelines, and needs no depth sensor (unlike [6, 5]).

The texture requirement. DA2 requiring textured terrain reflects monocular depth estimation assumptions, not a fundamental limitation. Real surfaces always have texture; textureless simulation is a rendering simplification.

Limitations. All results are simulation-only. We have deployed the PPO controller on a physical Go2, but the full student pipeline awaits additional domain augmentation. DA2-base limits training to 48 environments ($\sim$30 h vs. $\sim$2 h for the teacher). The ResNet-18 baseline achieves comparable sim performance with less compute; the domain-invariance advantage remains to be validated.

6 Conclusion

DA2 can produce canonical depth representations sufficient for quadruped parkour distillation, achieving 94–100% on stairs, 96–99% on hurdles, and within 7 pp of GT depth across gaps up to 0.50 m. Depth model capacity and terrain texture are the dominant factors; a DSQ metric explains the textureless failure mode. Future work: real-world deployment on Go2, DA2-large (300M), and dual-stream depth+RGB architectures.

A Reward Function Details

Table 5 lists all reward terms. Terrain scale $\beta = 0.35$ during stair/jump segments; 1.0 otherwise.

References

[1] Ananye Agarwal, Ashish Kumar, Jitendra Malik, and Deepak Pathak. Legged locomotion in challenging terrains using egocentric vision. In *Conference on robot learning*, pages 403–415. PMLR, 2023.

Table 5: Complete reward function.

Term	w_k	Expression	Term	w_k	Expression
Goal velocity	$+3.0$	$\exp(-\lVert v_{\text{des}} - v^{xy} \rVert^2/0.25)$	Collision	-2.0	$\sum_l \mathbb{1}[\lVert F_l \rVert > 0.1]$
Goal heading	$+1.5$	$\exp(-\theta_{\text{err}}^2/0.5)$	Stumble	-1.0	$\text{any}_f(\lVert F_f^{xy} \rVert > 4\lvert F_f^z \rvert)$
Progress	$+2.0$	$\text{clip}(\dot{p}_t, -0.5, 1.5)$	z-velocity	-0.5	$v_z^2 \cdot \beta$
Waypoint	$+20$	$\mathbb{1}[\text{reached}]$	Roll/pitch	-0.05	$(\omega_x^2 + \omega_y^2) \cdot \beta$
Success	$+40$	$\mathbb{1}[\text{completed}]$	Orientation	-0.5	$(g_x^2 + g_y^2) \cdot \beta$
Air time	$+1.0$	$\sum_f (t_{\text{air},f} - 0.3) \cdot \mathbb{1}[c_f]$	Flat back	-3.0	$w_{\text{asym}}(g_x - 0.09)^2 \cdot \beta$
Termination	-50	$\mathbb{1}[\text{non-timeout}]$	Torques	-2×10^{-4}	$\sum_j \tau_j^2$
Edge	-2.0	$\sum_f \text{edge}(p_f) \cdot c_f$	Act. rate	-0.01	$\lVert a_t - a_{t-1} \rVert^2$
Power	-2×10^{-4}	$\sum_j \lvert \tau_j \dot{q}_j \rvert$	Smoothness	-0.01	$\lVert a_t - 2a_{t-1} + a_{t-2} \rVert^2$
Accel.	-2×10^{-7}	$\sum_j \ddot{q}_j^2$	Joint lim.	-2.0	$\sum_j \max(0, \lvert q_j \rvert - q_j^{\text{lim}})$

[2] Guillaume Bellegarda, Chuong Nguyen, and Quan Nguyen. Robust quadruped jumping via deep reinforcement learning. *arXiv preprint arXiv:2011.07089*, 2020.

[3] Guillaume Bellegarda, Yiyu Chen, Zhuochen Liu, and Quan Nguyen. Robust high-speed running for quadruped robots via deep reinforcement learning. In *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2022.

[4] Ken Caluwaerts, Atil Iscen, J Chase Kew, Wenhao Yu, Tingnan Zhang, Daniel Freeman, Kuang-Huei Lee, Lisa Lee, Stefano Saliceti, Vincent Zhuang, et al. Barkour: Benchmarking animal-level agility with quadruped robots. *arXiv preprint arXiv:2305.14654*, 2023.

[5] Elliot Chane-Sane, Joseph Amigo, Thomas Flayols, Ludovic Righetti, and Nicolas Mansard. Soloparkour: Constrained reinforcement learning for visual locomotion from privileged experience. In *Proceedings of The 8th Conference on Robot Learning*, volume 270 of *Proceedings of Machine Learning Research*, pages 4268–4285. PMLR, 2025.

[6] Xuxin Cheng, Kexin Shi, Ananye Agarwal, and Deepak Pathak. Extreme parkour with legged robots. In *IEEE International Conference on Robotics and Automation (ICRA)*, 2024.

[7] Kyunghyun Cho, Bart van Merriënboer, Caglar Gulcehre, Dzmitry Bahdanau, Fethi Bougares, Holger Schwenk, and Yoshua Bengio. Learning phrase representations using RNN encoder-decoder for statistical machine translation. In *Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2014.

[8] Zipeng Fu, Ashish Kumar, Jitendra Malik, and Deepak Pathak. Minimizing energy consumption leads to the emergence of gaits in legged robots. In *Conference on Robot Learning (CoRL)*, 2021.

[9] Kaiming He, Xiangyu Zhang, Shaoqing Ren, and Jian Sun. Deep residual learning for image recognition. In *IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2016.

[10] Tairan He, Chong Zhang, Wenli Xiao, Guanqi He, Changliu Liu, and Guanya Shi. Agile but safe: Learning collision-free high-speed legged locomotion. *arXiv preprint arXiv:2401.17583*, 2024.

[11] David Hoeller, Nikita Rudin, Dhionis Sako, and Marco Hutter. Anymal parkour: Learning agile navigation for quadrupedal robots. *Science Robotics*, 2024.

[12] Shintaro Inoue, Kento Kawaharazuka, Kei Okada, and Masayuki Inaba. Body design and gait generation of chair-type asymmetrical tripedal low-rigidity robot. In *2024 IEEE 7th International Conference on Soft Robotics (RoboSoft)*, pages 593–600. IEEE, 2024.

[13] Diederik P. Kingma and Jimmy Ba. Adam: A method for stochastic optimization. In *International Conference on Learning Representations (ICLR)*, 2015.

[14] Ashish Kumar, Zipeng Fu, Deepak Pathak, and Jitendra Malik. RMA: Rapid motor adaptation for legged robots. In *Robotics: Science and Systems (RSS)*, 2021.

[15] Joonho Lee, Jemin Hwangbo, Lorenz Wellhausen, Vladlen Koltun, and Marco Hutter. Learning quadrupedal locomotion over challenging terrain. *Science robotics*, 5(47):eabc5986, 2020.

[16] Zhongyu Li, Xue Bin Peng, Pieter Abbeel, Sergey Levine, Glen Berseth, and Koushil Sreenath. Robust and versatile bipedal jumping control through multi-task reinforcement learning. *arXiv preprint arXiv:2302.09450*, 1, 2023.

[17] Yecheng Jason Ma, William Liang, Hungju Wang, Sam Wang, Yuke Zhu, Linxi Fan, Osbert Bastani, and Dinesh Jayaraman. Dreureka: Language model guided sim-to-real transfer. In *Robotics: Science and Systems (RSS)*, 2024.

[18] Gabriel B Margolis, Tao Chen, Kartik Paigwar, Xiang Fu, Donghyun Kim, Sangbae Kim, and Pulkit Agrawal. Learning to jump from pixels. *arXiv preprint arXiv:2110.15344*, 2021.

[19] Maxime Oquab, Timothée Darcet, Théo Moutakanni, et al. DINOv2: Learning robust visual features without supervision, 2023.

[20] René Ranftl, Katrin Lasinger, David Hafner, Konrad Schindler, and Vladlen Koltun. Towards robust monocular depth estimation: Mixing datasets for zero-shot cross-dataset transfer. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 44(3):1623–1637, 2022. doi: 10.1109/TPAMI.2020.3019967.

[21] Stéphane Ross, Geoffrey J. Gordon, and Drew Bagnell. A reduction of imitation learning and structured prediction to no-regret online learning. In *Proceedings of the Fourteenth International Conference on Artificial Intelligence and Statistics*, pages 627–635. JMLR Workshop and Conference Proceedings, 2011.

[22] Nikita Rudin, David Hoeller, Philipp Reist, and Marco Hutter. Learning to walk in minutes using massively parallel deep reinforcement learning. In *Conference on Robot Learning (CoRL)*, pages 91–100. PMLR, 2022.

[23] John Schulman, Philipp Moritz, Sergey Levine, Michael I. Jordan, and Pieter Abbeel. High-dimensional continuous control using generalized advantage estimation. In *International Conference on Learning Representations (ICLR)*, 2016.

[24] John Schulman, Filip Wolski, Prafulla Dhariwal, Alec Radford, and Oleg Klimov. Proximal policy optimization algorithms, 2017.

[25] Josh Tobin, Rachel Fong, Alex Ray, Jonas Schneider, Wojciech Zaremba, and Pieter Abbeel. Domain randomization for transferring deep neural networks from simulation to the real world. In *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2017.

[26] Lihe Yang, Bingyi Kang, Zilong Huang, Zhen Zhao, Xiaogang Xu, Jiashi Feng, and Hengshuang Zhao. Depth anything V2, 2024.

[27] Alan Yu, Ge Yang, Ran Choi, Yajvan Ravan, John J. Leonard, and Phillip Isola. Learning visual parkour from generated images, 2024.

[28] Ziwen Zhuang, Zipeng Fu, Jianren Wang, Christopher Atkeson, Sören Schwertfeger, Chelsea Finn, and Hang Zhao. Robot parkour learning. In *Conference on Robot Learning (CoRL)*, 2023.